CONTACT-deekshagoswami2743@gmail.com

PIZZA SALES

A SQL PROJECT

-BY DEEKSHA GOSWAMI

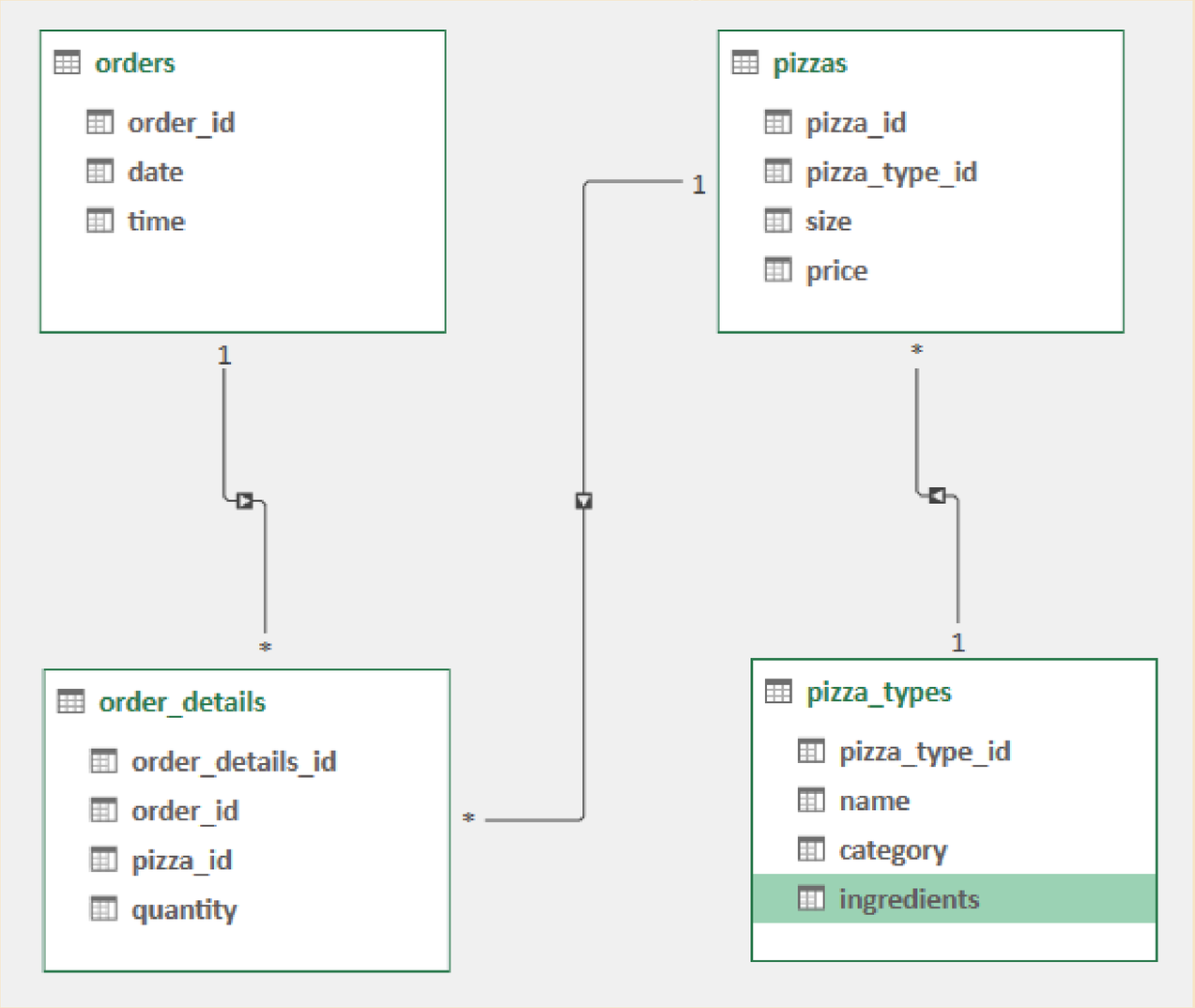
ABOUT PROJECT

This project analyzes a pizza sales dataset using SQL to extract meaningful business insights. It showcases the use of key SQL concepts such as SELECT, WHERE, JOIN (INNER JOIN), GROUP BY, and ORDER BY, along with aggregate functions like COUNT, SUM, and AVG to summarize data.

Advanced techniques including subqueries, CTEs, and window functions (RANK, cumulative SUM) are used to solve complex problems like revenue contribution, top-performing pizzas, and sales trends over time. The project emphasizes strong query logic, efficient data handling, and clear structuring to transform raw data into actionable insights.

TABLE STRUCTURES

This SQL project uses 4 tables to analyze data



Questions solved

Basic:

Q1- Retrieve the total number of orders placed.

Q2- the total revenue generated from pizza sales.

Q3-Identify the highest-priced pizza.

Q4-Identify the most common pizza size ordered.

Q5-List the top 5 most ordered pizza types along with their quantities.

Intermediate:

Q6-Join the necessary tables to find the total quantity of each pizza category ordered.

Q7-Determine the distribution of orders by hour of the day.

Q8-Join relevant tables to find the category-wise distribution of pizzas.

Q9- the orders by date and calculate the average number of pizzas ordered per day.

Q10-Determine the top 3 most ordered pizza types based on revenue.

Advanced:

Q11-Calculate the percentage contribution of each pizza type to total revenue.

Q1- Retrieve the total number of orders placed

QUERY RUN-

```
-- Retrieve the total number of order placed.

select count(order_id) as total_orders from orders;
```

RESULT-

Result Grid

	total_orders
▶	21350

Q2-The total revenue generated from pizza sales.

QUERY RUN-

```
-- Calculate the total revenue generated from pizza sales.

select sum(orderdetails.quantity * pizzas.price) as total_revenue
from orderdetails join pizzas
on orderdetails.pizza_id = pizzas.pizza_id;
```

RESULT-

Result Grid

total_revenue
817860.049999993

Q3-Identify the highest-priced pizza..

QUERY RUN-

```
-- Identify the highest-priced pizza.

select pizza_types.name,pizzas.price
from pizza_types join pizzas
on
pizza_types.pizza_type_id = pizzas.pizza_type_id
order by price desc limit 1;
```

RESULT-

Result Grid | Filter Rows

name	price
The Greek Pizza	35.95

Q4-. Identify the most common pizza size ordered.

QUERY RUN-

```
-- Identify the most common pizza size ordered.

select count(orderdetails.pizza_id),pizzas.size
from orderdetails join pizzas
on
orderdetails.pizza_id=pizzas.pizza_id
group by size;
```

RESULT-

Result Grid | Filter Rows:

count(orderdetails.pizza_id)	size
15385	M
18526	L
14137	S
544	XL
28	XXL

Q5- List the top 5 most ordered pizza types along with their quantities.

QUERY RUN-

```
-- List the top 5 most ordered pizza types along with their quantities.

SELECT pizzas.pizza_type_id,sum(orderdetails.quantity) as quantity
from pizzas join orderdetails
on pizzas.pizza_id=orderdetails.pizza_id
group by pizzas.pizza_type_id order by quantity desc limit 5;
```

RESULT-

pizza_type_id	quantity
classic_dlx	2453
bbq_ckn	2432
hawaiian	2422
pepperoni	2418
thai_ckn	2371

Q6- Join the necessary tables to find the total quantity of each pizza category ordered.

QUERY RUN-

```
select sum(orderdetails.quantity) as total_quantity, pizza_types.category
from orderdetails join pizzas on
orderdetails.pizza_id=pizzas.pizza_id
join pizza_types
on pizza_types.pizza_type_id=pizzas.pizza_type_id
group by pizza_types.category order by total_quantity;
```

RESULT-

Result Grid | Filter Rows:

total_quantity	category
11050	Chicken
11649	Veggie
11987	Supreme
14888	Classic

Q7- Determine the distribution of orders by hour of the day.

QUERY RUN-

```
-- Determine the distribution of orders by hour of the day.

select hour(order_time) as hours , count(orders.order_id)
from orders group by hours;
```

RESULT-

hours	count(orders.order_id)
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
0	1

Q8- Join relevant tables to find the category-wise distribution of pizzas.

QUERY RUN-

```sql
-- Join relevant tables to find the category-wise distribution of pizzas.

select category,count(name) from pizza_types
group by category;

```

RESULT-

Result Grid | Filter Rov

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Q9- **Group the orders by date and calculate the average number of pizzas ordered per day.**

QUERY RUN-

```sql
-- Group the orders by date and calculate the average number of pizzas ordered per day.
select avg(total_quantity) from
(select orders.order_date,sum(orderdetails.quantity) as total_quantity
from orders join orderdetails
on orders.order_id=orderdetails.order_id
group by orders.order_date) as order_quantity ;
```

RESULT-

Result Grid

	avg(total_quantity)
▶	138.4749

Q10- Determine the top 3 most ordered pizza types based on revenue.

QUERY RUN-

```
select pizza_types.name,sum(pizzas.price*orderdetails.quantity) as revenue
from pizza_types join pizzas on
pizza_types.pizza_type_id=pizzas.pizza_type_id
join orderdetails on
pizzas.pizza_id=orderdetails.pizza_id
group by pizza_types.name
order by revenue desc limit 3 ;
```

RESULT-

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Q10- Calculate the percentage contribution of each pizza type to total revenue.

QUERY RUN-

```
select pizza_types.name,sum(pizzas.price*orderdetails.quantity*100)/817860.049999993 as per_contri
from pizza_types join pizzas on
pizza_types.pizza_type_id=pizzas.pizza_type_id
join orderdetails on
pizzas.pizza_id=orderdetails.pizza_id
group by pizza_types.name
order by per_contri desc ;
```

RESULT-

name	per_contri
The Thai Chicken Pizza	5.310719113863108
The Barbecue Chicken Pizza	5.2292565213327595
The California Chicken Pizza	5.063152308270878
The Classic Deluxe Pizza	4.668341484585331
The Spicy Italian Pizza	4.258827656394306
The Southwest Chicken Pizza	4.243482732773205
The Italian Supreme Pizza	4.093212524563376
The Hawaiian Pizza	3.9460602092008625
The Four Cheese Pizza	3.945137068377442
The Sicilian Pizza	3.7831044565632306
The Pepperoni Pizza	3.687886454412373
The Greek Pizza	3.4790915634038178
The Mexicana Pizza	3.2744905439506713
The Five Cheese Pizza	3.1871589766489055
The Pepper Salami Pizza	3.121438686239806
The Italian Capocollo Pizza	3.068251102373837
The Vegetables + Vegetable...	2.9803081835333822
The Prosciutto and Arugula ...	2.9581161226789607
The Napolitana Pizza	2.945124902481813
The Spinach and Feta Pizza	2.8453828989446546
The Big Meat Pizza	2.8083044281231486
The Pepperoni, Mushroom	2.3020001111088708

name	per_contri
The Italian Vegetables Pizza	1.9586786272297
The Calabrese Pizza	1.948285651071982
The Spinach Pesto Pizza	1.9069277194796512
The Mediterranean Pizza	1.8781330620073853
The Spinach Supreme Pizza	1.8680151940420775
The Green Garden Pizza	1.7063738472126277
The Brie Carre Pizza	1.4169294612201806

CONCLUSION

This project demonstrates how SQL can be effectively used to analyze real-world business data and generate actionable insights. By applying a combination of fundamental and advanced SQL techniques, the analysis uncovers key trends in sales, customer preferences, and revenue performance.

It highlights the importance of structured querying, logical problem-solving, and data-driven decision-making. Overall, the project reflects strong proficiency in SQL and showcases the ability to turn raw data into valuable insights that can support business strategy and growth.

THANK YOU